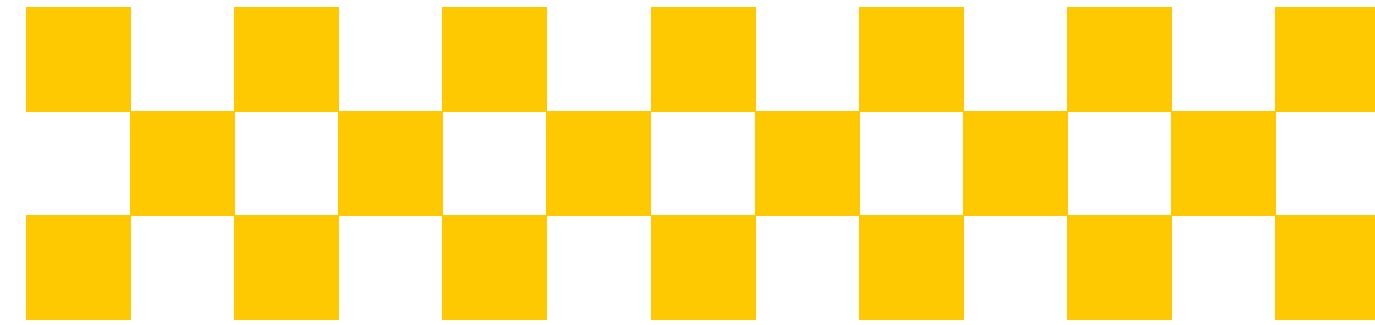


NYC TAXI DEMAND PREDICTION

Ashley Stevens, Harshita Ahuja, Brooke Stevens

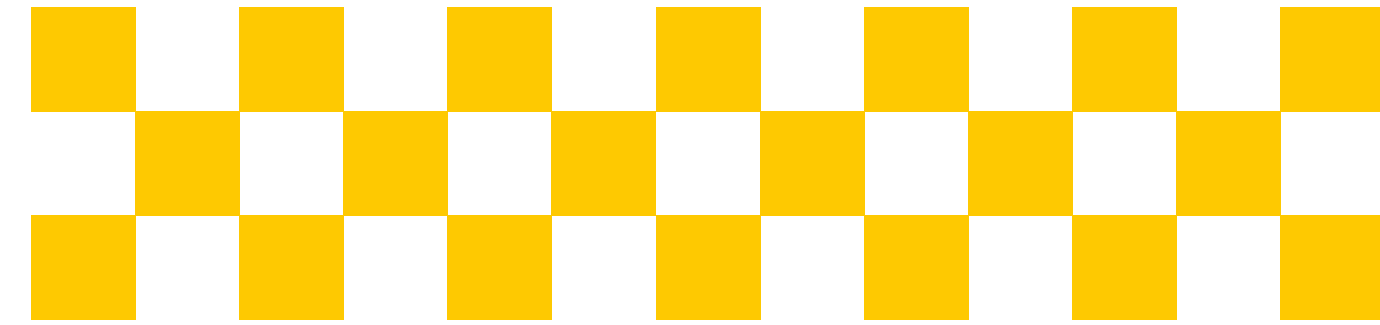


PROBLEM & MOTIVATION



- Predicts hourly taxi pickup demand
- **How many pickups will occur in a given location and hour?**
- Why this matters:
 - Better driver positioning
 - Reduced passenger wait times
 - Improved planning during rush hour
- The company dataset is big – we need Spark!

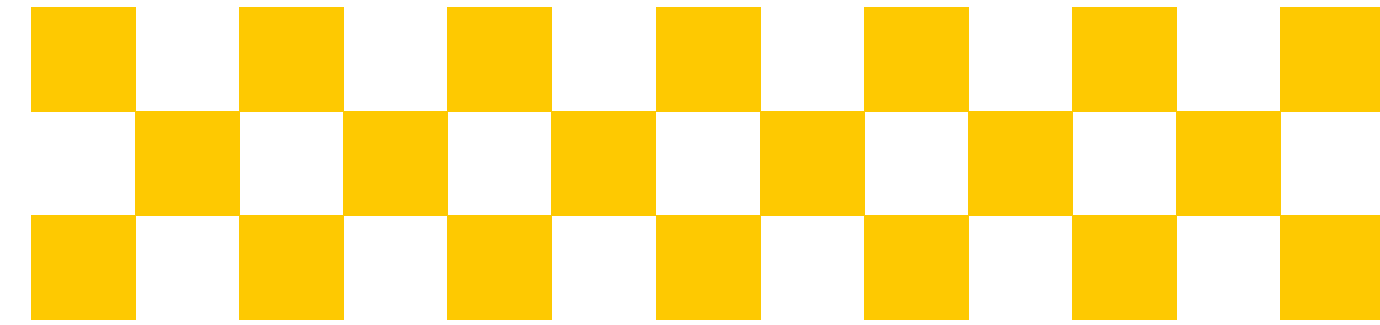
DATASET



- NYC Yellow Taxi Trip Data, January 2015
- 1.99 GB, contains 12.75 million rows/trips
- Key features:
 - Pickup/dropoff timestamps
 - Pickup/dropoff coordinates
 - Passenger count
 - Trip distance
 - Fare and tip amount

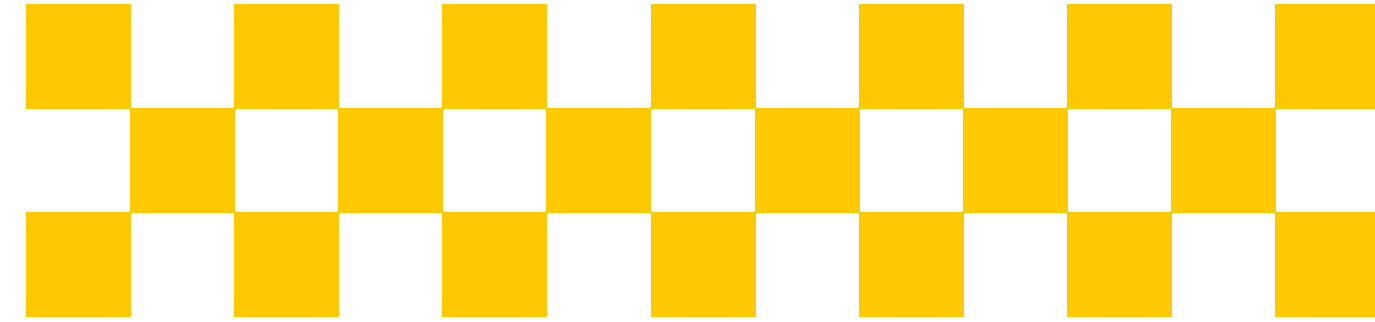


DATASET CHALLENGES



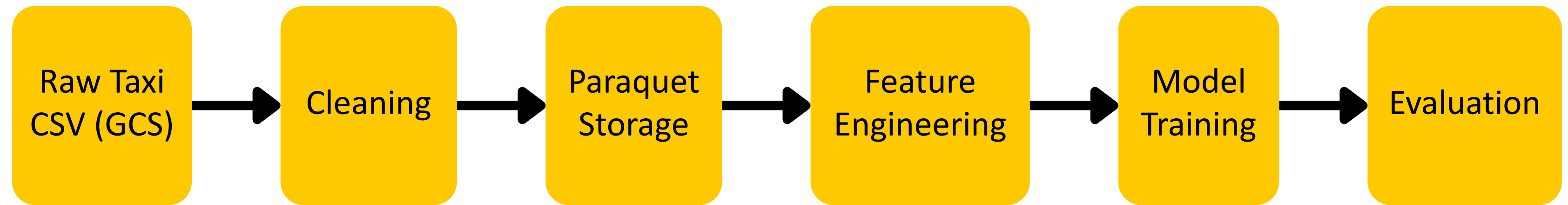
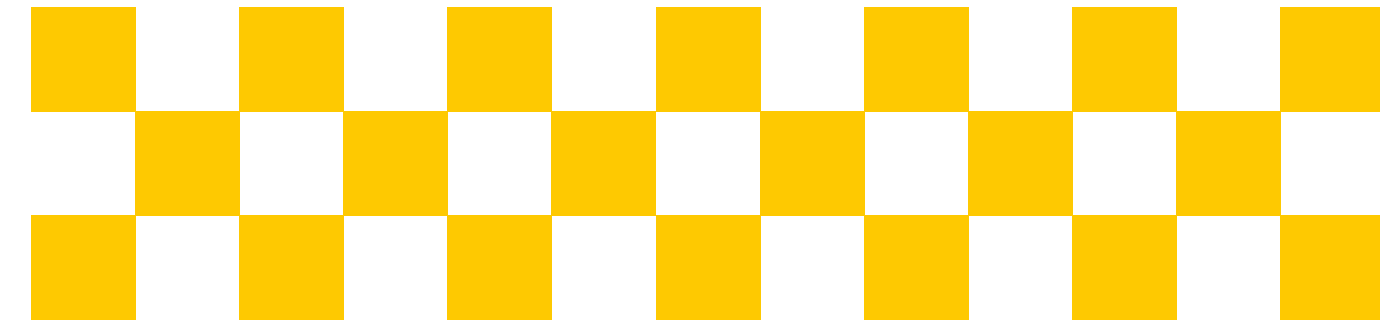
- Removed trips with:
 - Zero/negative distances and fares
 - <1 passenger or >6 passengers
 - Duration <1 minute or >3 hours
 - Average speed >100 mph
 - Missing pickup/dropoff times
 - Pickup/dropoff coordinates outside a NYC bounding box we defined

TASK & APPROACH



- **Task: Predict hourly pickup count for each spatial grid cell**
 - Aggregated trip-level data by hour
 - Engineered temporal, spatial, trip-level, and historical demand features
 - Compared 3 Spark MLlib regression models:
 - Linear Regression (baseline model)
 - Decision Tree
 - Random Forest

SYSTEM / PIPELINE

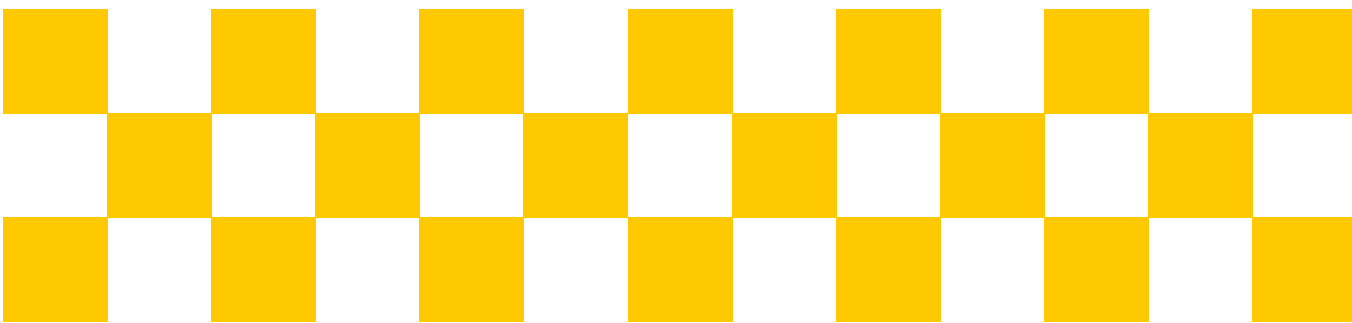


Tools used:

- GCS for storage
- Google Cloud Dataproc for distributed Spark processing
- PySpark in JupyterLab
- Parquet for faster downstream reads



RESULTS

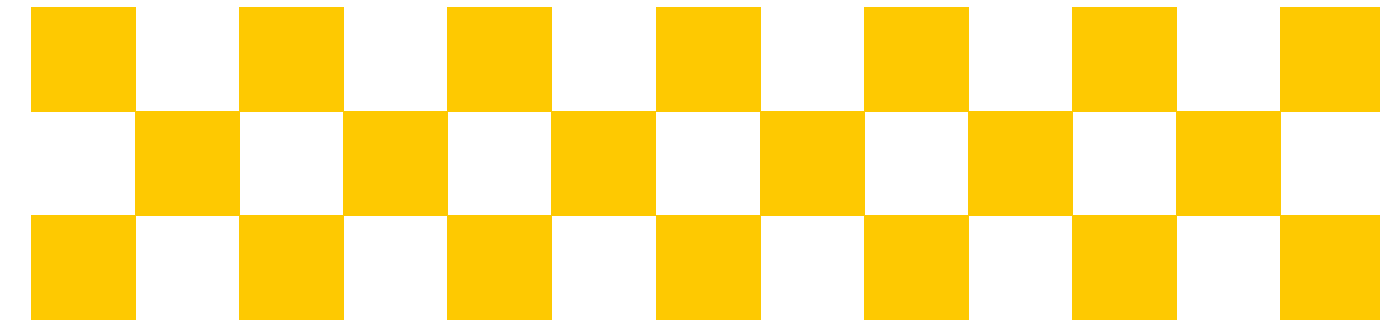


	Linear Regression (baseline)	Decision Tree	Random Forest
RMSE	46.93	43.57	32.83
MAE	19.74	15.57	12.98



Nonlinear models performed best

RESULTS

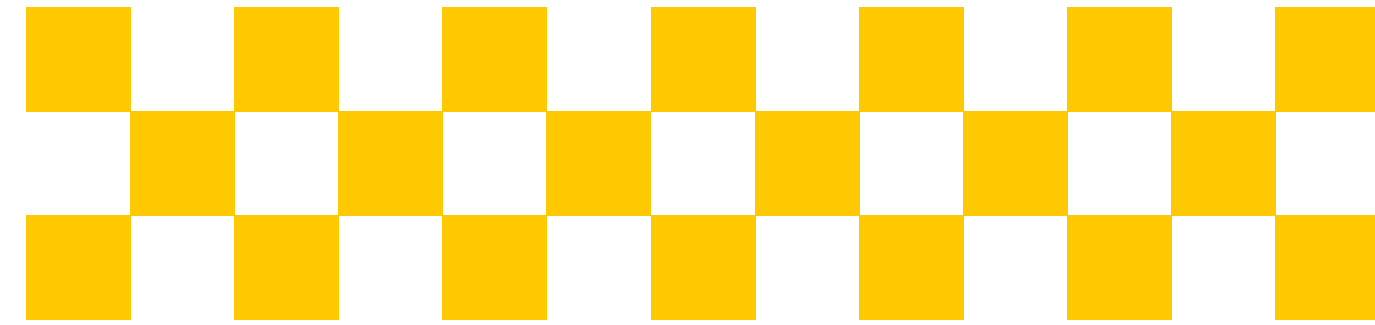


	Linear Regression (baseline)	Decision Tree	Random Forest
Training Time	3.7 s	8.1 s	70.2 s

70.2 s is highlighted with an orange rounded rectangle and a clock icon below it.

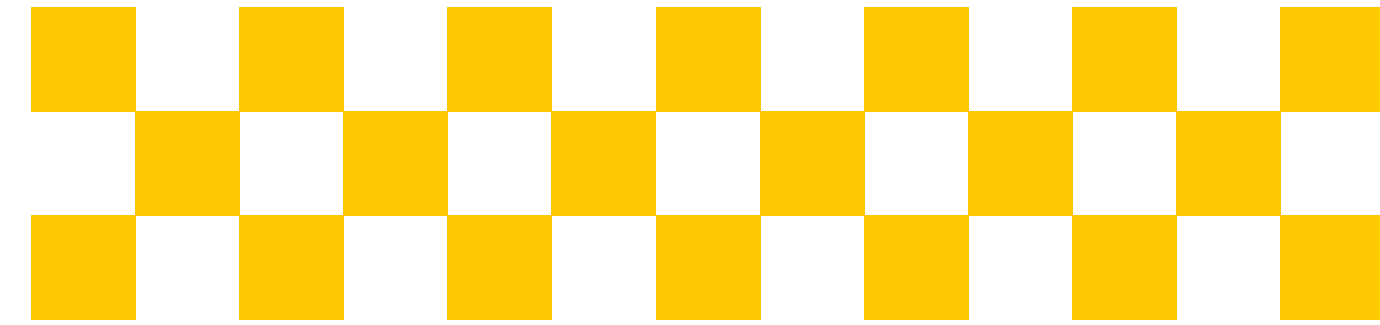
Tradeoff between performance and training time

CHALLENGES & LIMITATIONS

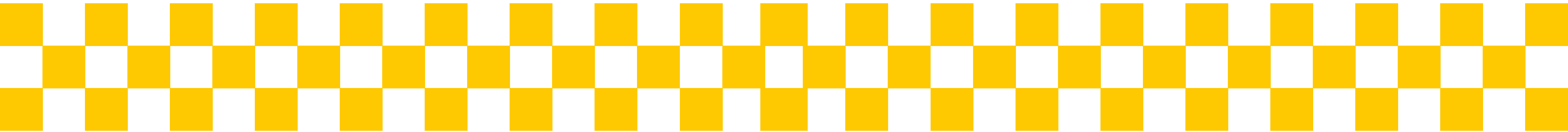


- Not enough compute left to try more complex models like Gradient Boosted Trees
- Only used 1 month of data (January 2015) so may not generalize to other months/seasons
- Doesn't take events or holidays into account
- NYC bounding box was a basic latitude-logitude rectangle
- Spatial grid cell bucketing reduced geographic precision

FUTURE WORK



- Using other meaningful signals like weather, holiday, or event data
- Exploring more complex models that require more compute (GBT)
- Using additional months of data
- Using more specific/precise city bounding and spatial representation
- Incorporating streaming for real-time data



THANK YOU!

